

PYTHON COMPUTING LAB

IV Semester

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6AE13	ESC	L	T	P	C	CIA	SEE	Total
		-	--	2	1	40	60	100

LIST OF PYTHON PROGRAMS

1. Write a program to demonstrate different number data types in Python.

2. Write a program to perform different Arithmetic Operations on numbers in Python.

3. Write a program to create, concatenate and print a string and accessing sub-string from a given string.

4. Write a python script to print the current date in the following format "Sun Oct 30 02:26:23 IST 2022"

5. Write a program to create, append, and remove lists in python.

6. Write a program to demonstrate working with tuples in python.

7. Write a program to demonstrate working with dictionaries in python.

8. Write a python program to find largest of three numbers.

8. Write a program to perform equations of uniform motion of kinematics :

i. $v = u + at$

ii. $s = ut + \frac{1}{2}(at^2)$

iii. $v^2 = u^2 - 2as$

9. Write the menu program to find the relationship between stress and strain curve as given below:

i. Young's Modulus

ii. Shear Modulus

iii. Poisson Ratio

10. Write the program to determine the shear force and bending moment in beams.

11. Write a program to calculate total pressure exerted in ideal fluid as equation is given below:

$P + \frac{1}{2}(\rho v^2) + \rho gh = \text{constant}$

Where P is Pressure, V is Velocity of fluid, ρ is density and h is the height of the container.

12. Write a program to compute eigen value and vector of a given 3*3 matrix using NumPy.

Note: Minimum 12 experiments are to be conducted.

Reference Books:

1. Reema Thareja, Python Programming: Using Problem Solving Approach, Oxford University Press.

2. Lutz M, Learning Python, Shroff

COURSE OUTCOMES : STUDENTS SHOULD BE ABLE TO:

1. Apply conditional statement, loops condition and functions in python program
2. Solve mathematical and mechanical problems using python program
3. Plot various type of chart using python program
4. Analyze the mechanical problem using python program
5. Illustrate programs on various python libraries such as Numpy, pandas and matplotlib